Numerical Algorithm for Non-Newtonian Blood Flow Simulation

1. **initialization of the variables and memory allocation based on the staggered grid approach**

Grid Resolution: 1400×35 cells (0.1mm resolution)

2. **Numerical Methodology**

The simulation employs a **pressure-based finite volume method** with staggered grid arrangement (MAC scheme):

Staggered Grid Features:

- **Spatial Discretization**: 2nd-order central differencing for diffusion terms
- **Convection Scheme**: QUICK (Quadratic Upstream Interpolation) for stability
- **Temporal Discretization**: Explicit Euler method with adaptive time stepping

3. **SIMPLE Algorithm Workflow**

Initialize flow field (u, v, p, μ)

Apply boundary conditions

Calculate initial shear rate ($\dot{\gamma}$) **and** viscosity (μ)

while t < t_max:

Momentum Prediction

Solve x-momentum → u*

Solve y-momentum → v*

Pressure-Velocity Coupling

Calculate divergence: $\nabla\cdot(u^*,v^*)$

Solve pressure Poisson equation: $\nabla^2 p = \nabla\cdot(u^*,v^*)/\Delta t$

Velocity Correction

Correct velocities: $u = u^* - \Delta t/\rho\ \nabla p$

$v = v^* - \Delta t/\rho\ \nabla p$

Viscosity Update

Recalculate shear rate **and** μ using Bird-Carreau model

Adaptive Time Stepping

Δt = CFL * min(Δx,Δy)/max(|u|,|v|)

Output

if time_step % output_interval == 0:

Save VTK files **and** CSV data

4. Boundary Conditions

Boundary	**u-velocity**	**v-velocity**	**Pressure**
Inlet	Parabolic profile	No-slip	Zero gradient
Outlet	Zero gradient	No-slip	Fixed (0 Pa)
Walls	No-slip (u=0)	No-slip (v=0)	Zero gradient

5. Stability Measures

1. **CFL Condition**:
2. **Viscosity Clamping**:
   $\mu \in [\mu_{\infty}, \mu_0]$ to prevent unrealistic values
3. **Residual Monitoring**:
   Convergence check for pressure solver (L_2 norm < $1e^{-5}$)

6. Validation Metrics

python

Divergence-Free Check: $\|\nabla \cdot u\|_2 < 1e^{-6}$

Flow Rate Conservation: |Q_in - Q_out| < 1%

7. Implementation Details

- **Hardware Acceleration**: CUDA implementation for GPU parallelism
- **Visualization**: ParaView-compatible VTK output with:
  - Velocity vectors
  - Pressure contours
  - Shear rate distribution